convert that grid into ciphertext differs from the route cipher technique. The process is as follows:

1. We first select a keyword. Let the selected keyword be "DUFFER".
2. Now, we remove the repeating letters from the keyword. Now, the keyword becomes "DUFER".
3. Next we re-arrange the letters of the keyword we obtained in the previous step. It becomes: "DEFRU".
4. We give each letter a number in sequence. D -> 1, E -> 2, F -> 3, R -> 4, U -> 5.
5. Now we take the keyword from step 2 and replace the letters with numbers we got in step 4 (last step). So the keyword now becomes "1 5 3 2 4".
6. We now lay out a table with 5 columns. The number sequence we obtained in the last step are placed in the top row serving as heads.
7. We then lay out our message in the table linearly. The table would look like:

| 1 | 5 | 3 | 2 | 4 |
|---|---|---|---|---|
| I | L | O | V | E |
| D | I | G | I | T |
| M | A | G | A | Z |
| I | N | E | X | X |

   Please do note that 'X' in the last row serves as padding to avoid irregularities in the encryption. When decrypting the ciphertext on the other end, it would be easily detectable and thus can be discarded.
8. Next we write the text in the column number 1 (from top to bottom) followed by text in column number 2 (again from top to bottom). The ciphertext comes out as: "IDMIVIAXOGGEETZXLIAN".

The keyword "DUFFER" can still be used to decrypt the message. Try doing that yourself if you want to put your grey matter through its paces.

With that we round up the simple ciphers and will now try to figure out more modern techniques which provide higher protection than these classical and simple (and also weak) ciphers provide. Nonetheless they are important because they help us understand how we have progressed so far to a place and time where clever cryptanalysis techniques combined with superfast computers fail to get back the plaintext within any practically meaningful time limits.

## Modern Ciphers

We've spoken about some of the simple ciphers till now which worked by